

A Project Report for the Bachelor's in Computer Application

VELORA



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School of Environmental Science and Management
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Pokhara University, Nepal

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A Project Report for the Bachelor's in Computer Application

VELORA

Supervised by

Mr. Sumant Yadav

A project submitted in partial fulfilment of the requirements for the degree of project for the Bachelor in Computer Application.

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School of Environmental Science and Management

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July 2026

ORIGINAL COPY OF THE APPROVAL

This project entitled **VELORA: A Secure, Trust-Based Multi-Vendor E-Commerce Platform** prepared and submitted by **Ankita Baral, Sadikshya Chhetri, and Rashni Thapa Magar**, has been examined by us and is hereby accepted as partial fulfillment of the requirements for the degree of **Bachelor of Computer Application (BCA)** awarded by **Pokhara University**.

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CERTIFICATE OF AUTHENTICATED WORK

We hereby declare that this study entitled **VELORA: A Secure, Trust-Based Multi-Vendor E-Commerce Platform** is based on our original work. Related works on the topic by other researchers have been duly acknowledged. The project documentation in this report was mobilized by three below-mentioned sixth semester undergraduate students of Bachelor of Computer Application (BCA) for the partial fulfillment of the requirement for the project of the sixth semester under the supervision of **Sumant Yadav**.

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CERTIFICATE

RECOMMENDATION

This is to certify that this project entitled **VELORA: A Secure, Trust-Based Multi-Vendor E-Commerce Platform** Platform prepared and submitted by **Ankita Baral, Sadikshya Chhetri, and Rashni Thapa Magar**, in partial fulfillment of the requirements for the degree of Bachelor of Computer Application (BCA) awarded by Pokhara University, has been completed under my supervision. I certify that this work is original and has been carried out in accordance with the academic requirements of the University. I recommend this project for acceptance by Pokhara University.

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Sumant Yadav

(Supervisor)

School of Environmental Science and Management

Date:

ROLE AND RESPONSIBILITY FORM

Phase / Task	Responsibilities	Assigned To
Requirement Analysis	Gathering requirements, defining project scope, and preparing software/hardware requirements.	Rashni Thapa Magar, Ankita Baral & Sadikshya Chhetri
System & Database Design	Designing the system architecture, ER Diagram, Use Case Diagram, DFDs and UML Diagram.	Rashni Thapa Magar
Backend Development	Developing APIs, authentication, database integration, business logic and server-side functionalities.	Rashni Thapa Magar & Sadikshya Chhetri
Frontend Development	Developing responsive user interfaces, integrating APIs, and implementing application features.	Anita Baral & Sadikshya Chhetri
Testing & Debugging	Testing system functionalities, fixing bugs and validating application performance.	Rashni Thapa Magar, Ankita Baral & Sadikshya Chhetri
Documentation	Preparing the proposal, project report and user documentation	Rashni Thapa Magar, Ankita Baral & Sadikshya Chhetri

ABSTRACT

VELORA: A Secure, Trust-Based Multi-Vendor E-Commerce Platform is a web-based application developed to provide a secure, transparent, and trustworthy online marketplace by connecting buyers with verified sellers, unlike conventional e-commerce platforms that primarily rely on ratings and reviews. VELORA empathize with seller authenticity through administrator-approved seller verification, trust scores, verified badges, and detailed meet-the-seller profiles, enabling buyers to make informed purchasing decisions.

The platform supports secure reserve authentication using Email OTP verification, Google OAuth, and JWT-based authorization. Buyers can browse products, search by category, manage shopping carts and wish lists, place orders, track order status, submit verified reviews, and communicate with sellers through real-time messaging. Verified sellers can manage products, inventory, orders, coupons, etc., and shop profiles through a dedicated dashboard, while administrators oversee seller verification, user management, platform activities, and overall system operations.

VELORA is developed using React.js for the frontend, the Django REST framework for the backend, and SQLite3 as the database management system. Additional technologies, including Redux Toolkit, Redis, Django channels, and Tailwind CSS, enhance application performance, scalability, and user experience by integrating secure authentication, seller verification, transparency features, and modern e-commerce functionalities. VELORA provides a reliable multi-vendor marketplace that strengthens buyer confidence while supporting the growth of trusted local businesses.

Keywords: Multi-Vendor E-Commerce, Seller Verification, Seller Transparency, Buyer Trust, React.js, Design Goal Risk Framework, SQLite3, JWT Authentication, Google OAuth, Email OTP Verification, Real-Time Messaging, Redis, Project Management and Order Management

ACKNOWLEDGEMENT

All the members of the group have equally contributed to the completion of this project entitled **VELORA: A Secure, Trust-Based Multi-Vendor E-Commerce Platform**. Foremost, we would like to thank **Pokhara University** and **School of Environmental Science and Management** for assigning this type of project which would help in our future development. We are also thankful to **Sumant Yadav**, the project supervisor, for his guidance and support throughout the project.

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Thanking from

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LIST OF ABBREVIATIONS

API	Application Programming Interface
BCA	Bachelor's in Computer Application
CSS	Cascading Style Sheet
DB	Database
DFD	Data Flow Diagram
ER	Entity-Relationship
HTML	Hyper Text Markup Language
HTTP	Hypertext Transfer Protocol
JS	JavaScript
OOP	Object-Oriented Programming
ORM	Object Relational Mapper
OTP	One-Time Password
REST	Representational State Transfer
SQL	Structured Query Language
UI/UX	User Interface / User Experience
UML	Unified Modeling Language
URL	Uniform Resource Locator

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CHAPTER 1

INTRODUCTION

1.1 Background

The rapid advancement of internet technology has transformed the way people buy and sell products. Multi-vendor e-commerce platforms have made online shopping more convenient by allowing customers to purchase products from multiple sellers through a single marketplace. However, despite their popularity, establishing trust between buyers and sellers remains a major challenge due to unverified sellers, misleading product information, fake reviews, and limited communication.

VELORA is a trust-focused, multi-vendor e-commerce platform developed to address these challenges by promoting transparency, security, and seller authenticity. Every seller must register and submit verification documents before listing products, and only administrator-approved sellers are allowed to sell on the platform. Buyers can browse products as guests and after registering, can manage carts and wishlists, place orders using a dummy payment system, communicate with sellers, and submit verified product reviews. Seller trust is strengthened through review-based trust scores [1].

The platform also provides dedicated dashboards for sellers and administrators. Sellers can manage products, customer orders, and messages, while administrators oversee seller verification, users, products, and platform activities through a centralized dashboard. These features create a secure, transparent, and well-managed online marketplace for all users.

VELORA is developed using React.js, Tailwind.css, Redux Toolkit, and Axios for the frontend, while the backend is built with Python, Django, Django REST framework, Django Channels, and Simple JWT [2]. SQLite3 serves as the database with ready supporting catching and real-time messaging by integrating modern web technologies with trust-oriented features [3]. VELORA aims to provide a secure, reliable, and user-friendly online shopping experience.

1.2 Objectives

General Objective

To design and develop VELORA, a trust-focused, multi-vendor e-commerce platform that provides a secure, transparent, and user-friendly online marketplace for buyers, sellers, and administrators.

Specific Objectives

- To implement secure user registration and authentication using email OTP and Google OAuth.
- To provide role-based access for buyers, sellers, and administrators.
- To implement seller verification before product listing.
- To enable product browsing, cart, wish list, ordering, and verified reviews.
- To provide real-time messaging between buyers and sellers.
- To calculate seller trust scores based on verified reviews.
- To develop seller and administrator dashboards for platform management.
- To build a secure, scalable, and responsive web application using modern web technologies.

1.3 Purpose, Scope, and Applicability

1.3.1 Purpose

The primary purpose of VELORA is to develop a trust-focused, web-based, multi-vendor e-commerce platform that provides a secure, transparent, and reliable marketplace for buyers, sellers, and administrators. The system aims to strengthen buyer confidence by ensuring that only administrator-verified sellers can list products, while enabling buyers to make informed purchasing decisions through seller profile trust scores and verify the product reviews [4].

VELORA also aims to simplify online marketplace management by providing role-based access control, secure authentication through email OTP verification and Google OAuth, product management, shopping cart and wishlist management, order processing with a dummy payment system, real-time buyer-seller messaging, centralized administrative control. By integrating these features with modern web technologies, the platform delivers a responsive, secure, and user-friendly e-commerce solution [5].

1.3.2 Scope

What the system does:

- Provides a trust-focused multi-vendor marketplace connecting buyers with verified sellers.
- Supports secure user registration, email OTP verification, Google OAuth, and role-based access.
- Enables seller verification, product management, shopping cart, wish list, order placement, and verified reviews.
- Provides real-time buyer-seller messaging and trust score calculation.
- Includes seller and administrator dashboards for managing products, orders, users, and platform activities

What the system does NOT do:

- Does not integrate a real online payment gateway, dummy payment only.
- Does not provide delivery tracking or logistics management.
- Does not include AI-based product recommendations.
- Does not support multi-warehouse inventory management.
- Does not include a dedicated mobile application.

1.3.3 Applicability

VELORA Ecommerce is applicable to:

- **Online Buyers:** Users who want to purchase products from verified sellers with greater trust and confidence.
- **Local Sellers:** Businesses that want to showcase their products, build credibility, and reach more customers through a trusted platform.
- **Small Businesses and Startups:** Individuals looking for a secure marketplace to start and manage their online business.
- **Administrators:** Platform managers who need tools to verify sellers, manage users, monitor products, and maintain marketplace activities.

1.4 Achievements

The following achievements were successfully completed during the development of VELORA:

- **Trust-focused multi-vendor platform:** Developed a web-based marketplace connecting buyers with administrator-verified sellers.
- **Seller verification system:** Implemented an admin approval process with verification documents and seller trust scores to enhance buyer confidence.
- **Secure authentication:** Integrated Secure User Authentication with Email OTP Verification, Google OAuth, and Role-Based Access Control.
- **Complete e-commerce workflow:** Developed product browsing, product search, shopping cart, wishlist, order placement, dummy payment, and verified review features.
- **Real-time communication:** Implemented buyer-seller messaging for direct communication regarding products and orders.
- **Seller and administrator dashboards:** Develop dashboards for managing products, orders, users, seller verification, and overall platform activities.
- **Responsive user interface:** Designed a modern responsive interface using React.js and Tailwind CSS for an improved user experience.
- **Scalable system architecture:** Build the application using Django REST Framework, SQLite3, Redis, and modular architecture for maintainability and scalability.

1.5 Organization of Report

This report is organized into seven chapters. It's covering a specific aspect of the development of VELORA.

- **Chapter 1 – Introduction:** Presents the background, objectives, purpose, scope, applicability, achievements, and overall organization of the report
- **Chapter 2 – Survey of Technologies:** Reviews related e-commerce systems, relevant technologies and the tools and frameworks used in the development of VELORA.
- **Chapter 3 – Requirements and Analysis:** Describe the problem definition, system requirements, planning, and scheduling software and hardware

requirements, primarily product description and conceptual models, including Use Case Diagram, Data Flow Diagram (DFDs), ER Diagram and UML Class Diagram.

- **Chapter 4 – Design:** Explain the overall system design, database design, interface design, system architecture, and module design of the application.
- **Chapter 5 – Implementation and Testing:** Discusses the implementation approach, coding details, code efficiency, testing methods, test cases, and improvements made during system development.
- **Chapter 6 – Results and Discussion:** Presents the implementation results, test reports, system functionalities, and user documentation.
- **Chapter 7 – Conclusions:** Summarize the project, highlight the significance of the system, discuss its limitations, and outlines possible future enhancements.

CHAPTER 2

SURVEY OF TECHNOLOGIES/ LITERATURE REVIEW

2.1 Review of Similar/Relevant Projects

Before developing VELORA, several popular e-commerce platforms were studied to understand their features, strengths and limitations. This review helped identify gaps in seller transparency, trust management, and buyer-seller communication, which influenced the design of the proposed system [6].

Table 2.1: Comparison of Existing Multi-Vendor E-Commerce Platforms

Platform	Features Offered	Limitations
Amazon/Daraz	Multi-vendor marketplace, product search, shopping cart, secure checkout, customer reviews	Limited seller transparency, no mandatory seller verification, buyers rely mainly on platform reputation
Alibaba	Business-to-business marketplace, supplier profiles, buyer-seller messaging, quotation, requests, secure transactions	Supplier verification varies by membership level, trust depends in certifications and buyer reviews, and the platform primarily targets wholesale trading rather than general retail.
Etsy	Seller storefronts, product reviews, shop profiles, handmade and unique products	No formal seller verification, trust is based primarily on ratings and customer feedback.
Shopify/WooCommerce	Customizable online stores, inventory and order management	Primarily designed for single-vendor stores and lack built-in multi-vendor trust and verification mechanisms.

Existing platforms provide comprehensive e-commerce features but offer limited seller verification and transparency. VELORA overcomes this limitation through administrator-approved seller verification, trust scores, verified reviews, real-time messaging and dedicated dashboards for buyer-sellers and administrators.

CHAPTER 3

REQUIREMENTS AND ANALYSIS

3.1 Problem Definition

Although existing multi-vendor e-commerce platforms have made online shopping more convenient, they still face several challenges related to security, seller transparency, buyer confidence, and marketplace management. These challenges reduce user trust and affect the overall shopping experience. The key problems addressed by VELORA are:

1. **Lack of secure authentication and role-based access:** Many platforms provide limited authentication mechanisms and insufficient role-based control, affecting system security and user management.
2. **Lack of seller verification and transparency:** Buyers often cannot determine whether sellers are genuine because many platforms do not require mandatory verification or provide sufficient seller information.
3. **Limited trust and communication between buyers and sellers:** Existing platforms mainly rely on ratings and reviews while offering limited direct communication, making it difficult for buyers to make informed purchasing decisions.
4. **Inefficient marketplace management:** Buyers, sellers, and administrators require dedicated tools to manage products, orders, reviews, users, and other marketplace activities efficiently.
1. **5.Lack of an integrated trust-focused e-commerce platform:** Existing systems rarely combine secure authentication, seller verification, trust scores, verified reviews, buyer-seller messaging, and centralized management into a single platform.

To address these challenges, VELORA provides a trust-focused multi-vendor e-commerce platform featuring secure authentication, administrator-approved seller verification, role-based access control, real-time buyer-seller messaging, verified reviews, trust scores and dedicated dashboards for buyers, sellers, and administrators

3.2 Requirements Specification

The requirements of VELORA define the functional and non-functional capabilities needed to develop a secure, trust-focused multi-vendor e-commerce platform.

3.2.1 Functional Requirements

Functional requirements define the essential operations that the VELORA system shall perform for buyers, sellers, and administrators. The system shall:

- Allow users to register and log in using Email OTP Verification or Google OAuth.
- Provide role-based access for guest, buyers, sellers, and administrators.
- Allow guests to browse and search the products without logging in.
- Allow seller to register, submit verification documents, and list products only after administrator approval.
- Enable buyers to browse products, manage shopping carts and wishlists, and place orders using a dummy payment system.
- Display product details, seller information, trust scores, verified reviews, and seller profiles.
- Allow buyers and sellers to communicate through real-time messaging.
- Allow buyers to submit verified product reviews after successful purchases.
- Allow sellers to manage products, customer orders, and order status.
- Provide administrator task boards for managing users, seller verification, products, orders, reviews and overall platform activities.

3.2.2 Non-Functional Requirements

Non-functional requirements define the quality, security, and performance standards of VELORA. The system shall:

- Provide secure authentication using JWT, email OTP verification, and Google OAuth.
- Ensure role-based authentication and protection of user data.
- Deliver a responsive and user-friendly interface across the desktop and mobile devices.
- Maintain reliable performance with fast response times.
- Use a modular architecture to support scalability and future maintenance.
- Store and manage data securely using SQLite3.

- Support real-time messaging through Django channels and Redis.
- Allow future integration of features such as online payment gateways and mobile applications.

3.2.3 Requirements Prioritization Table

This table prioritizes the system requirements based on their importance to ensure that the core functionalities of VELORA are developed first.

Table 3.1: Requirement Prioritization Table

Requirement	Priority	Justification
User authentication (Email OTP and Google OAuth)	High	Required for secure user access
Role-Based Access Control	High	Ensures secure access for buyers, sellers and administrators.
Seller Verification and Approval	High	Core feature for building buyer trust.
Product Catalog, Search and Categories	High	Essential for product discovery.
Card, wishlist, and order placement.	High	Core shopping functionality.
Seller Dashboard	High	Enables sellers to manage products and orders.
Administrator Dashboard	High	Required for platform management and seller approval.
Real-Time Buyer Seller Messaging	Medium	Improves communication and customer experience
Verified Reviews and Trust Score	Medium	Enhances transparency and marketplace credibility.
Domain Payment System	Low	supports order placement during development without a real payment integration.

3.3 Planning and Scheduling

Before starting the development of the VELORA, a project plan was prepared to organize the development process and ensure timely completion of each phase. The project was divided into several stages, including planning, system design, implementation, testing, documentation and deployment.

Table 3.2: Project Scheduling

Phase	Task	Duration
Planning	Requirement analysis, literature review, and project planning	Week 1
System Design	UI/UX design, database design, UML diagrams, DFDs and ER diagram	Week 2
Backend Development	Authentication, REST APIs, business logic, database implementation, and seller verification	Week 3 - 5
Frontend Development	User interface development using React.js, Tailwind CSS, Redux Toolkit, and API integration	Week 4 - 7
Feature Integration	Cart, wishlist, messaging, reviews, dashboards and dummy payment integration	Week 6 - 7
Testing	Unit testing, integration testing, system testing and bug fixing	Week 8
Documentation	Report preparation, project review and final improvements	Week 9
Deployment	Final system deployment and project demonstration	Week 9

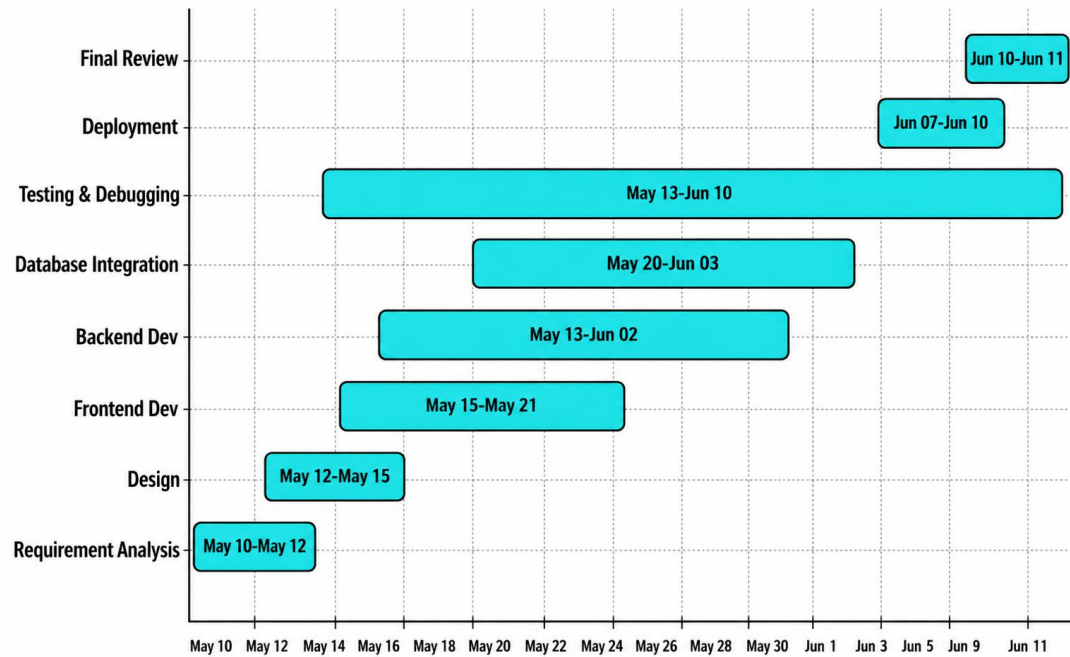


Figure 3.1: Gantt Chart

3.4 Software and Hardware Requirements

To develop and run VELORA, the following software and hardware requirements were identified.

3.4.1 Software Requirements

Table 3.3: Software Requirement

Component	Specification
Operating System	Windows 10/11, macOS, or Linux
Programming Language	Python 3.11+, JavaScript(ES6+)
Backend Framework	Django, Django REST Framework
Frontend Framework	React.js
Styling Framework	Tailwind CSS
State Management	Redux Toolkit

Component	Specification
HTTP Client	Axios
Database	SQLite3
Cache/Message Broker	Redis
Real-Time Communication	Django Channels(WebSocket)
Package Manager	Pip, npm
Code Editor	Visual Studio Code
Version Control	Git & GitHub
API Testing	Postman
Authentication	Django Simple JWT, Email OTP, Google OAuth
Design Tool	Figma

3.4.2 Hardware Requirements

Table 3.4: Hardware Requirement

Component	Specification
Processor	Dual-core 2.0 GHz or above
Memory	Minimum 4 GB RAM (8 GB recommended)
Storage	At least 10 GB free disk space
Network	Internet connection for API access, Email OTP, Google OAuth, and Real-Time Messaging
Display	Minimum resolution of 1280 × 720

3.4.3 Development Cost Estimation

The estimated development cost for **VELORA** is minimal because open-source technologies and personal devices were used throughout the project.

Table 3.5: Development Cost Estimation

Item	Estimated Cost (NPR)	Remarks
Development & Testing	Rs. 0	Performed on personal computers.
Software & Development Tools	Rs. 0	Open-source tools including Python, Flask, SQLite3, Tailwind CSS, Git, and Visual Studio Code.
Hosting & Deployment	Rs. 0	Application runs locally on the Flask development server and is not publicly deployed.
Documentation & Report Binding	Rs. 500	Printing, binding, and other documentation expenses.
Total Estimated Cost	Rs. 500	Estimated overall project development cost.

3.5 Preliminary Product Description

VELORA is a web-based, trust-focused multi-vendor e-commerce platform that follows a client-server architecture. The system consists of a React.js frontend that communicates with a Django REST framework backend through RESTful APIs, while SQLite3 stores application data and reads it. Supports sketching and real-time messaging.

Backend

The backend is developed using Python, Django, and Django REST framework. It manages user authentication, seller verification, product and category management, shopping cart, wishlist, orders, reviews, trust score calculation, buyer-seller messaging and administrative operations [7]. The system uses Django Simple JWT for authentication, Email OTP, and Google OAuth for secure user registration and login, and Django Channels with Redis to support real-time messaging.

Frontend

The frontend is developed using React.js, Tailwind CSS, Redux, Toolkit, and Axios. It provides a responsive and user-friendly interface for browsing products, viewing seller information and rush discos, managing shopping carts and wishlists, placing orders, communicating with sellers and accessing dedicated dashboards for buyers, sellers, and administrators [8].

3.6 Conceptual Models

The conceptual model provides a high-level representation of the proposed VELORA system by illustrating the major users, entities, and interactions required to fulfill the system requirements. It serves as a bridge between the requirements analysis and the system design by presenting the overall structure and functionality of the application without focusing on implementation details [9].

The conceptual model identifies the main functional components of the system, including user authentication, role-based access control, seller verification, product management, shopping cart and wish list management, order processing, buyer-seller messaging, review management, trust score calculation and administrative management. These components work together to provide a secure, transparent, and user-friendly multi-vendor e-commerce platform that enhances buyer confidence while supporting verified sellers.

CHAPTER 4:

DESIGN

4.1 Introduction

The design phase transforms the system requirements identified in the previous chapter into a detailed blueprint for implementation. It defines the overall architecture, database structure, system components, user interfaces, and interactions required to develop VELORA. This system is designed using a modular client-server architecture that separates the frontend, backend, and database layers, improving scalability, maintainability, and security.

To visualize the system structure and behavior, several design modules are used, including the Use Case Diagram, Data Flow Diagram (DFDs), Entity Relationship (ER) Diagram, UML Class Diagram and System Architectural Diagram. These modules provide a clear understanding of how different components of the system interact and how data flows throughout the application.

4.2 System Design

The system design illustrates the functional and structural aspects of VELORA through various UML and data flow models. These models provide a clear representation of user interactions, system processes, and the overall architecture before implementation.

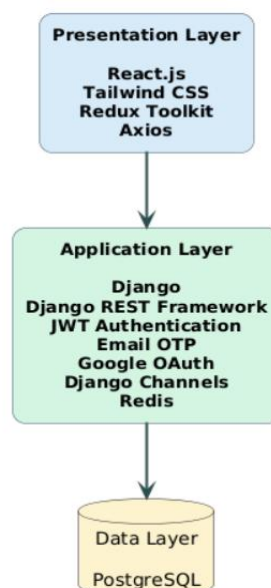


Figure 4.1: Technology Stack of VELORA

4.2.1 Use Case Diagram

The Use Case Diagram illustrates the interactions between the four primary actors of the VELORA system: Guest, Buyer, Seller, and Administrator. It identifies the major functionalities available to each user role and defines the system boundary. Guests can browse products without authentication, while buyers can manage shopping activities such as cart, wishlist, order, review, and messaging. Sellers can manage products, orders, and verification requests, whereas administrators oversee seller verification and overall platform management.

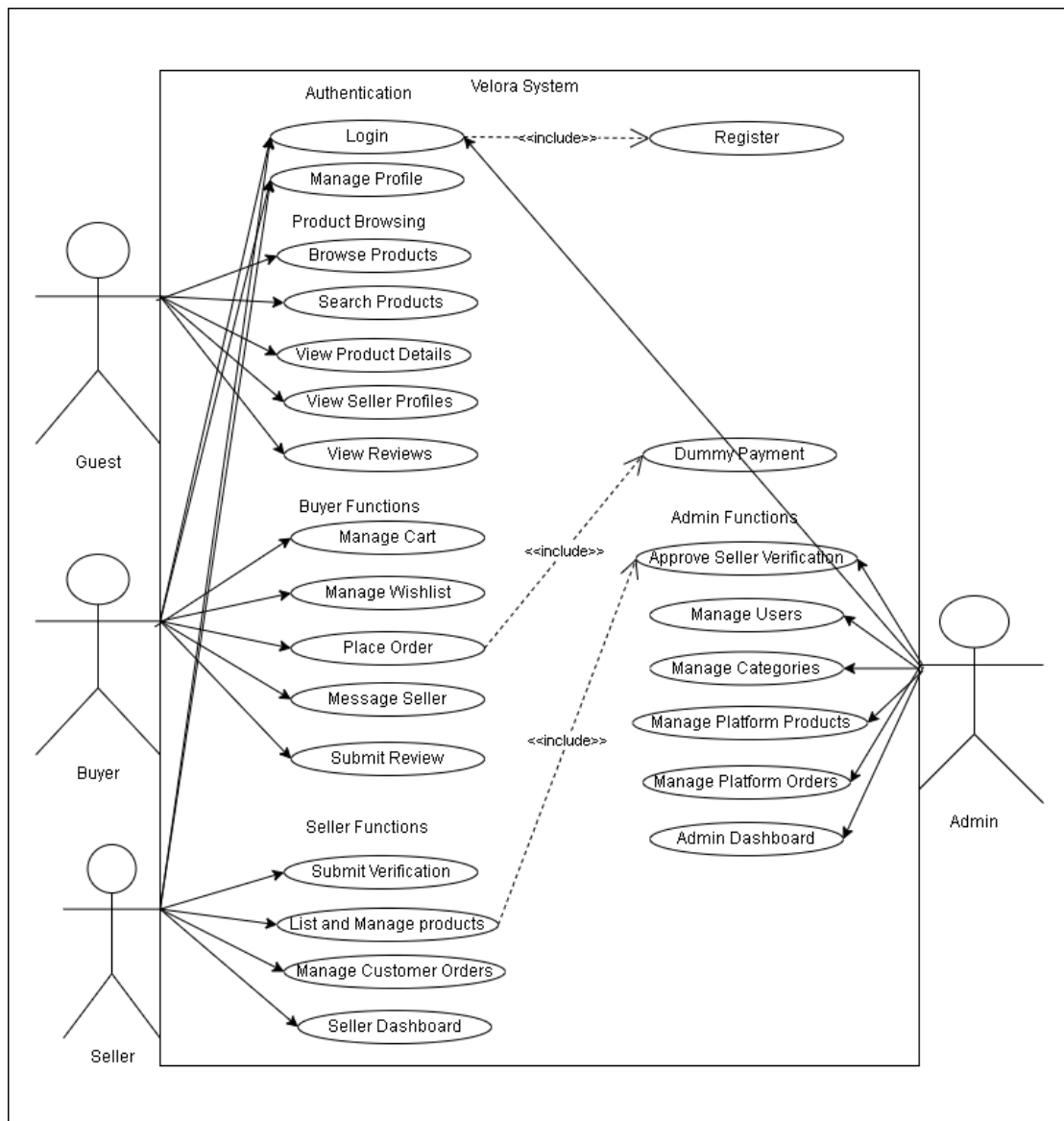


Figure 4.2: Use Case Diagram of VELORA

4.2.2 Data Flow Diagram (Level 0)

The Level 0 Data Flow Diagram provides a high-level overview of the VELORA system, by illustrating the flow of data between external entities and the system. It shows how Guests, Buyers, Sellers, and Administrators interact with the platform through major functions such as authentication, product browsing, order management, messaging, and administrative operations, without revealing internal processing details.

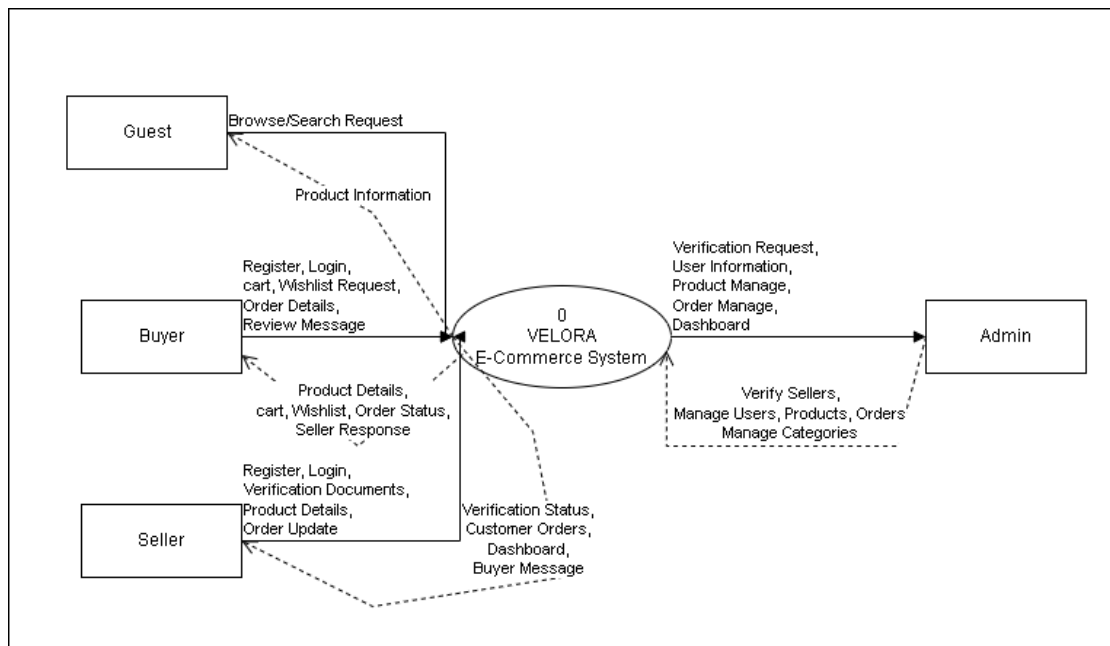


Figure 4.3: Data Flow Diagram (Level 0)

4.2.3 Data Flow Diagram (Level 1)

The Level 1 Data Flow Diagram expands the Level 0 model by illustrating the major internal processes of the VELORA system. It shows how data flows through authentication, seller verification, product management, shopping cart and wish list management, order processing, messaging, reviews and administrative functions providing a clear understanding of the system's internal workflow.

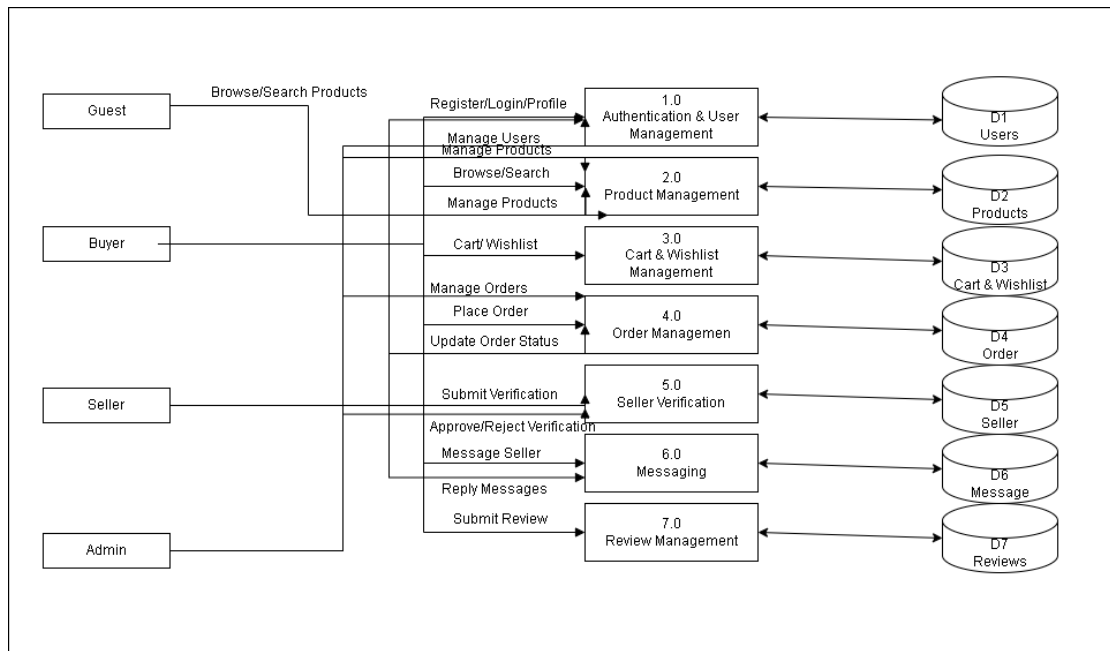


Figure 4.4: Data Flow Diagram (Level 1)

4.3 Database Design

The database design defines how data is organized, stored and managed within VELORA. A relational database model is used to maintain data consistency, reduce redundancy and establish relationships among different system entities. PostgreSQL is used as the database management system due to its reliability, security and scalability, making it suitable for managing the platform's data efficiently.

4.3.1 Entity Relationship (ER) Diagram

The Entity Relationship (ER) Diagram illustrates the logical structure of the VELORA database by showing the major entities and the relationships between them. It provides a clear representation of how data is organized and connected within the system, serving as the foundation for database implementation.

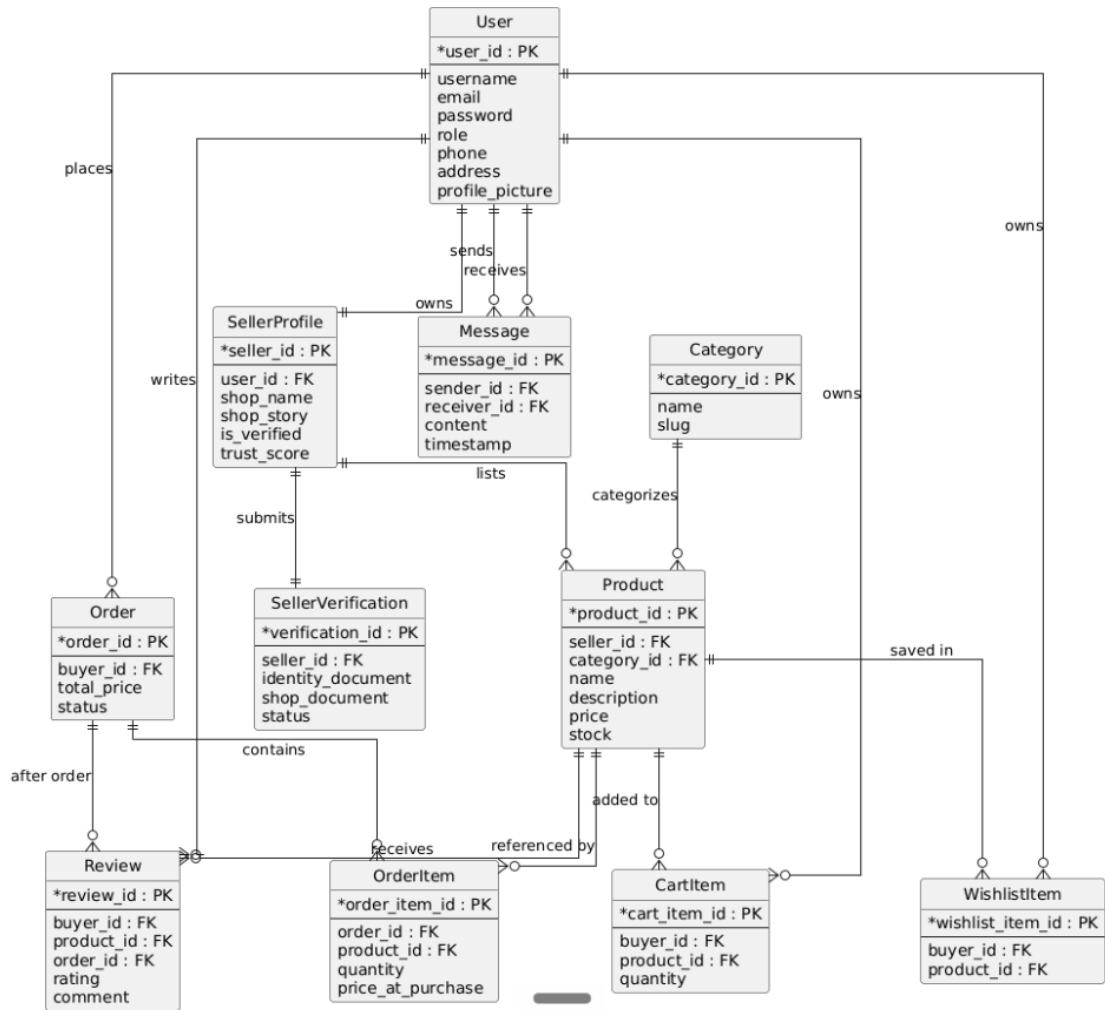


Figure 4.5: Entity Relationship (ER) Diagram of VELORA

4.4 Interface Design

The VELORA interface follows a modern, responsive, and user-friendly design across the following key screens:

- Home Page: Displays featured products, categories, search functionality, and navigation.
- Product Listing and Details: Allows users to browse products, view seller information, trust scores, and reviews
- Shopping Cart and Wishlist: Enable buyers to manage selected and saved products.
- CheckOut and Order Page: Displays the order summary and dummy payment option.

- Buyer Dashboard: Allows buyers to manage orders, reviews, wishlists, and messages.
- Seller Dashboard: Enables sellers to manage products, orders, and verification status.
- Administrator Dashboard: Allows administrators to manage users, sellers, products, orders, and platform activities.

4.4.1 UML Class Diagram

The UML Class Diagram represents the static structure of the VELORA system by illustrating the major classes, their attributes, methods, and relationships. It serves as a blueprint for the object-oriented implementation of the application.

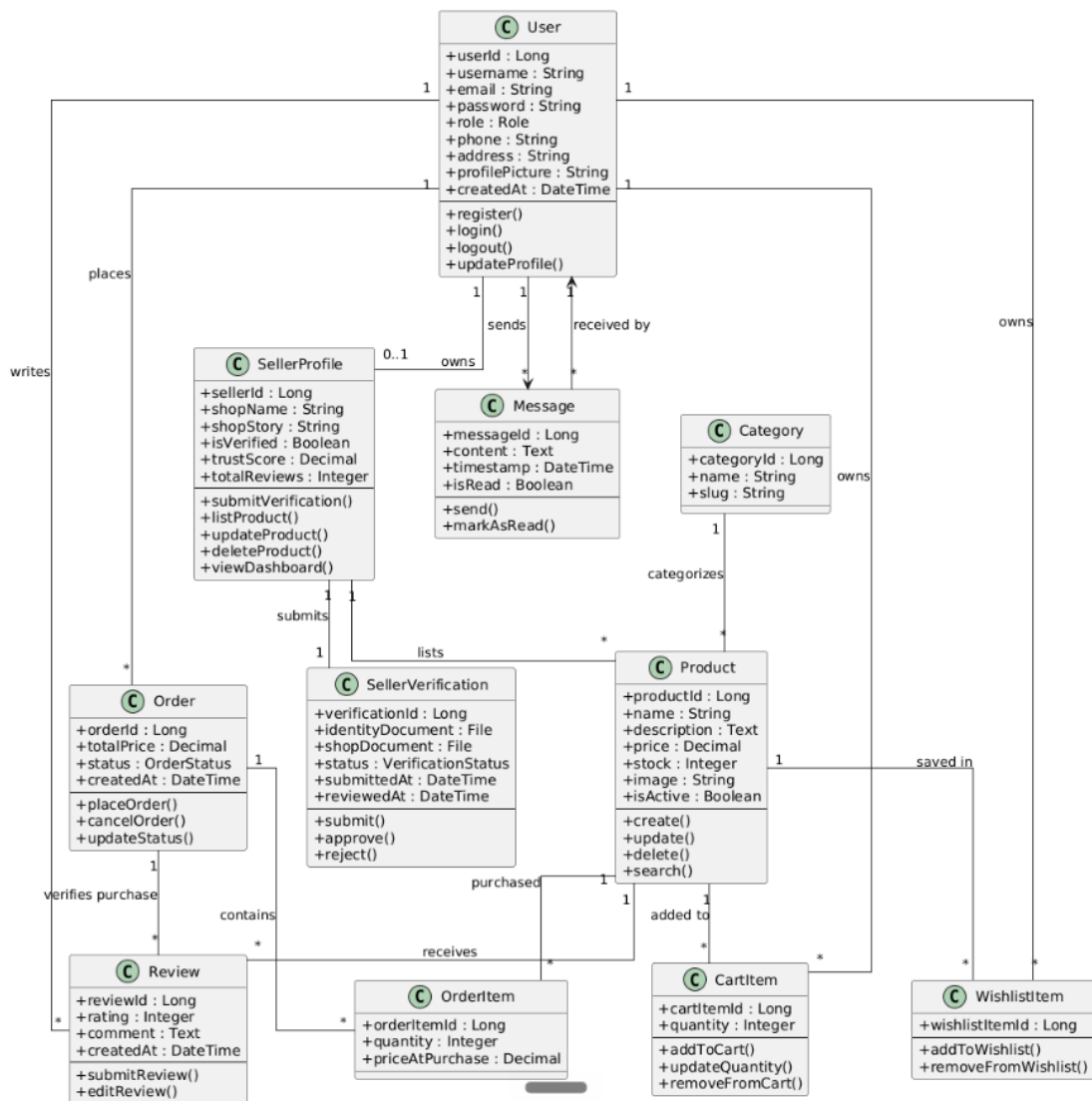


Figure 4.6: UML Class Diagram of VELORA

4.5 Summary

This chapter presented the design of the VELORA system, including the system models, database design, interface design, and UML Class Diagram. These design artifacts provide a clear blueprint for implementing the application and serve as the foundation for the implementation and testing phases discussed in the next chapter.

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1 Implementation Approaches

VELORA is implemented using a React.js frontend and a Django REST framework backend following a client-server architecture. The frontend communicates with the backend through RESTful APIs, while SQLite3 stores application data. This separation of concerns improves scalability, maintainability, and overall system performance [10].

The implementation consists of the following major components:

- **Frontend:** Developed using React.js, Tailwind CSS, Redux Toolkit, and Axios to provide a responsive and interactive user interface.
- **Backend:** Developed using Django and Django REST framework, implementing business logic, REST APIs, authentication, seller verification, order processing, and administrative operations.
- **Database:** SQLite3 is used to store user accounts, seller information, products, categories, orders, reviews, wishlists and messages.
- **Authentication:** Secure authentication is implemented using JWT, Email OTP verification, and Google OAuth.
- **Real-time Messaging:** Buyer-seller communication is implemented using Django Channels and Redis, enabling instant message delivery [11].

5.2 Coding Details and Code Efficiency

Two representative implementation features demonstrate the core functionality of this system.

1. **Seller Verification Workflow:** When a user registers as a seller, the system requires verification documents before allowing product listings. The administrator reviews the submitted information and approves or rejects the request. Only verified sellers are permitted to publish products, improving buyer trust and marketplace transparency.
2. **JWT of Indication and Role-based Access:** The system uses JSON Web Tokens (JWT) to authenticate users after successful login. Based on the

authenticated user's role (Buyer, Seller, or Administrator), appropriate permissions and dashboard access are provided, ensuring secure access to protected resources [12].

5.2.1 Code Efficiency

- **RESTful APIs:** Efficient API endpoints reduce unnecessary data transfer between the frontend and backend.
- **Redux Toolkit:** Centralized state management minimizes redundant API requests and improves application performance.
- **Database Optimization:** SQLite3 relationships and indexing improve query performance for products, orders, and users.
- **Real-Time Communication:** Redis and Django Channels enable efficient message delivery without continuous page refreshes.
- **Modular Architecture:** The project follows a modular structure, making the code easier to maintain, extend and test.

5.3 Testing Approach

Testing was performed at multiple levels to ensure the correctness and reliability of VELORA's core functionalities, including authentication, seller verification, product management, ordering, and user interactions. Each module was manually tested to verify proper system behavior before final integration [13].

5.3.1 Unit Testing

Individual functions related to database operations, OTP verification, image handling, coupon calculation, and order processing were tested separately to ensure each component worked correctly.

5.3.2 Integration Testing

- **Registration + Authentication:** Verified that user registration, Email OTP verification, Google OAuth and JWT authentication work together correctly.

- **Order Workflow:** Verified that adding products to the cart, placing orders using the dummy payment system and updating order status functions as expected.
- **Seller Verification:** Verified that approved sellers can list products while unverified sellers cannot.
- **Messaging System:** verified successful real-time communication between buyers and sellers.

5.3.3 Beta Testing

The system was tested by a small group of users representing buyers and sellers. Feedback was collected regarding usability, navigation, responsiveness and system performance. Minor interface improvements, validation enhancements and bug fixes were implemented based on the feedback received [14].

5.4 Modifications and Improvements

During development and testing, several improvements were made to enhance system performance, security and usability.

- Improved form validation for user registration and product management.
- Enhanced seller verification workflow and administrator controls.
- Optimized database queries to improve product search performance.
- Improved the responsiveness of the user interface across different screen sizes.
- Refined the real-time messaging functionality and notification handling.
- Improved error handling and validation throughout the application [15].

5.5 Test Cases

Table 5.1: Test Case Table

Test Case ID	Feature	Input	Expected Output	Result
TC-01	User Registration	Valid user details	Account created successfully	Passed
TC-02	Email OTP Verification	Correct OTP	User account verified	Passed
TC-03	Google OAuth Login	Valid Google account	User logged into the system	Passed
TC-04	Seller Verification	Valid verification documents	Seller approved by administrator	Passed
TC-05	Product Browsing	Search Keyword/Category	Matching products displayed	Passed
TC-06	Add to Cart	Select product	Product added to shopping cart	Passed
TC-07	Place Order	Cart items + Dummy Payment	Order successfully created	Passed
TC-08	Buyer-Seller Messaging	Send message	Message delivered successfully	Passed
TC-09	Product Review	Verified purchase review	Review submitted successfully	Passed
TC-10	Administrator Management	Manage users/products	Review submitted successfully	Passed

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Test Reports

The testing results presented in Chapter 5 demonstrate the VELORA successfully performs its core functionalities and meets the specified system requirements. The application provides a secure, transparent and user-friendly environment for buyers, sellers and administrators. The key outcomes are summarized below:

- **Secure Authentication:** User registration, Email OTP verification, Google OAuth and JWT based authentication ensure secure account access.
- **Seller Verification:** Sellers can list products only after administrator approval, improving marketplace trust and reliability.
- **Marketplace Operations:** Buyers can browse products, manage shopping carts and wishlists, place orders using the dummy payment system and submit product reviews successfully.
- **Real-Time Communication:** Buyers and sellers can communicate through the integrated messaging system, improving customer interaction.
- **Administrative Management:** The administrator dashboard effectively manages users, seller verification, products, orders, reviews and overall platform activities.

6.2 User Documentation

Buyer Walkthrough

A buyer registers using Email OTP verification or Google OAuth and logs into the system. After authentication, the buyer can browse products, search and filter products by category, and view detailed product information, including seller profiles, trust scores and verified reviews. Buyers can add products to their shopping cart or wishlist, proceed to checkout using dummy payment system and place orders. They can also view their order history, communicate with sellers through the messaging system, and submit product reviews after completing a purchase.

Seller Walkthrough

A seller registers for an account and submits the required verification information. Until the administrator approves the verification request, the seller cannot publish products. Once verified, the seller gains access to the seller dashboard, where they can add, update and remove products, manage customer orders, update order status, response to buyer messages, and monitor their shop activities through a centralized interface.

Admin Walkthrough

The admin logs into the system through the administrator account and access the admin dashboard. From the dashboard, the administrator reviews seller verification requests, approves or rejects sellers, manage users, products, categories, orders, reviews and oversees overall marketplace activities. The admin is responsible for maintaining the security, transparency and smooth operation of the VELORA platform.

CHAPTER 7

CONCLUSIONS

7.1 Conclusion

The VELORA project successfully demonstrates the development of a trust-focused multi-vendor e-commerce platform that connect buyers and verified sellers through a secure and transparent online marketplace. By utilizing react.js, Django REST Framework and SQLite3, the system provides secure authentication, seller verification, product management, order processing, real-time messaging and role-based access control. Trust-enhancing features such as seller verification, trust scores and verified reviews improve buyer confidence while enabling sellers to establish credibility. Overall, VELORA provides a scalable, user-friendly and reliable platform that addresses the limitations of traditional online marketplaces while meeting the objectives of the project.

7.1.1 Significance of the System

- **Trust-Focused Marketplace:** Builds buyer confidence through seller verification, trust scores and verified reviews.
- **Secure User Authentication:** Enhances account security using Email OTP verification, Google OAuth and JWT authentication.
- **Efficient Marketplace Management:** Provides dedicated dashboards for buyers, sellers and administrators to simplify platform operations.
- **Scalable System Architecture:** Uses modern web technologies to support future enhancements and long-term maintainability.

7.2 Limitations of the System

- **Dummy Payment System:** The current implementation uses simulated payment methods instead of real payment gateway integration.
- **Local development Environment:** The application is developed and tested locally and is not deployed to a production server.
- **Email-Based Verification:** User verification currently relies on Email OTP, without SMS-based verification.

- **Limited Recommendation Features:** Personalized product recommendations and AI-based suggestions are not included in the current version.

7.3 Future Scope of the Project

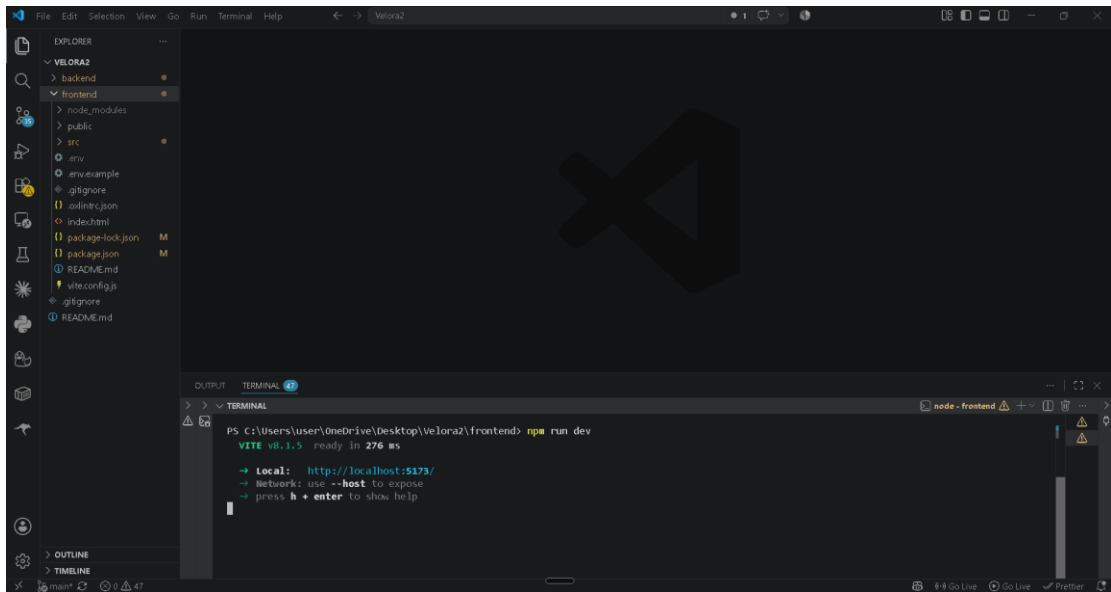
- **Real Payment Gateway Integration:** Integrate payment gateways such as Khalti, eSewa or Strip for secure online transactions.
- **AI-Based Product Recommendations:** Recommend products based on user preferences and purchase history.
- **Advanced Search and Filtering:** Enhance product discovery with intelligent search and filtering techniques.
- **Mobile Application:** Develop dedicated Android and iOS applications for improved accessibility.
- **Analytics Dashboard:** Provide sellers and administrators with detailed sales, customer, and marketplace analytics.
- **Cloud Deployment:** Deploy the platform to a cloud environment for improved scalability, reliability and availability.

References

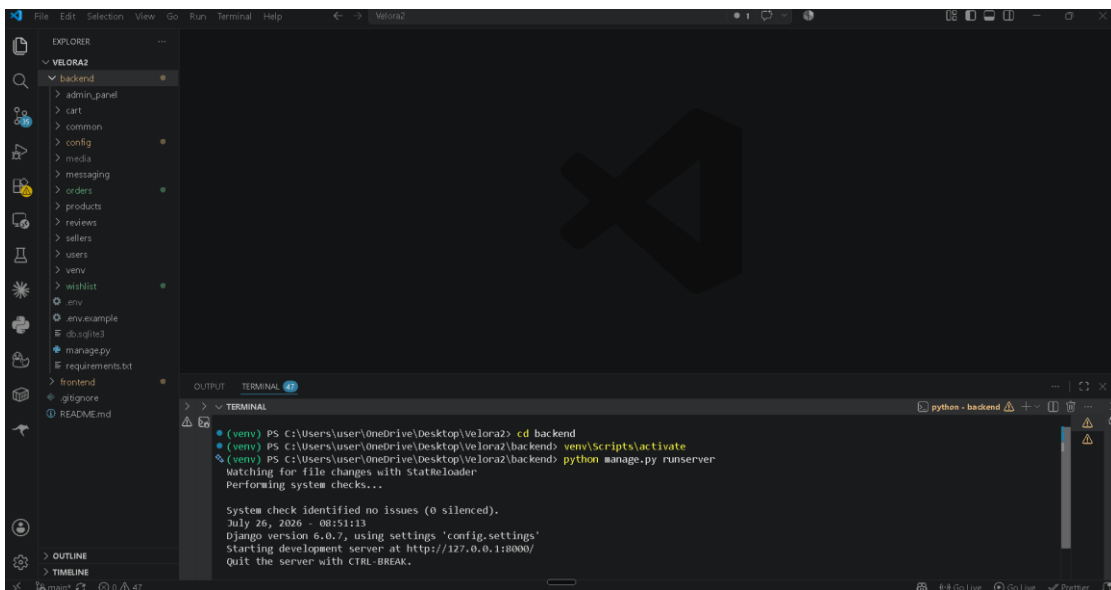
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APPENDICES



Appendix 1: Executing Frontend (React) on the Terminal



Appendix 2: Executing Backend (Django) on the Terminal


Home Log in SIGN UP

Create your account

Join the trusted local marketplace.

FULL NAME
Sadikshya chhetri

EMAIL
sadikshya.spark5@gmail.com

PASSWORD

I WANT TO
Shop as a buyer

CREATE ACCOUNT

Already have an account? [Log in](#)

OR

 Continue with Google

Appendix 1:Registration Page (OTP / Google OAuth)

Free delivery on orders over Rs. 50 • Verified sellers only


Home Log in SIGN UP

USD • EN

Welcome back

Log in to continue shopping or selling.

EMAIL
sadikshya.spark5@gmail.com

PASSWORD

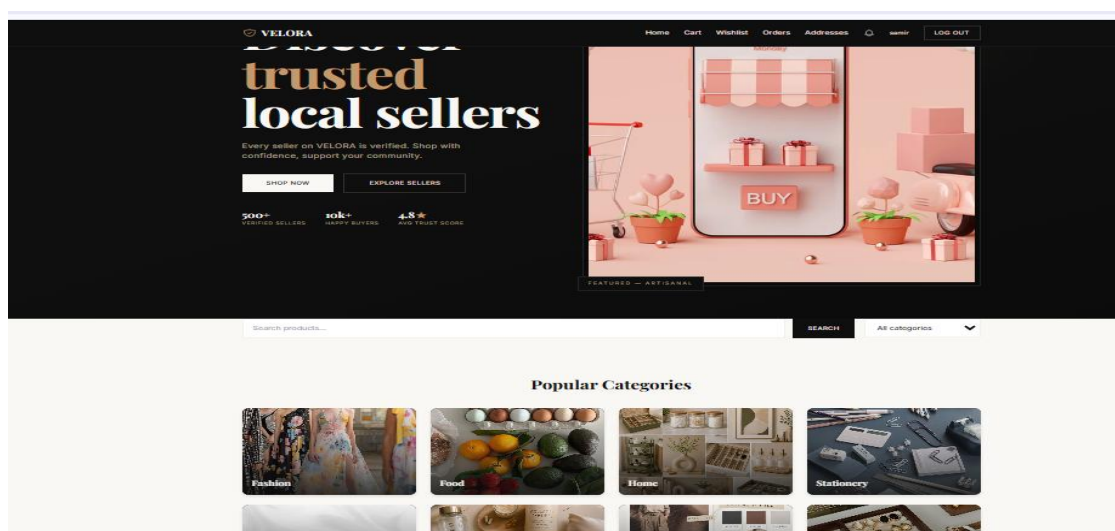
LOG IN

Don't have an account? [Create one](#)

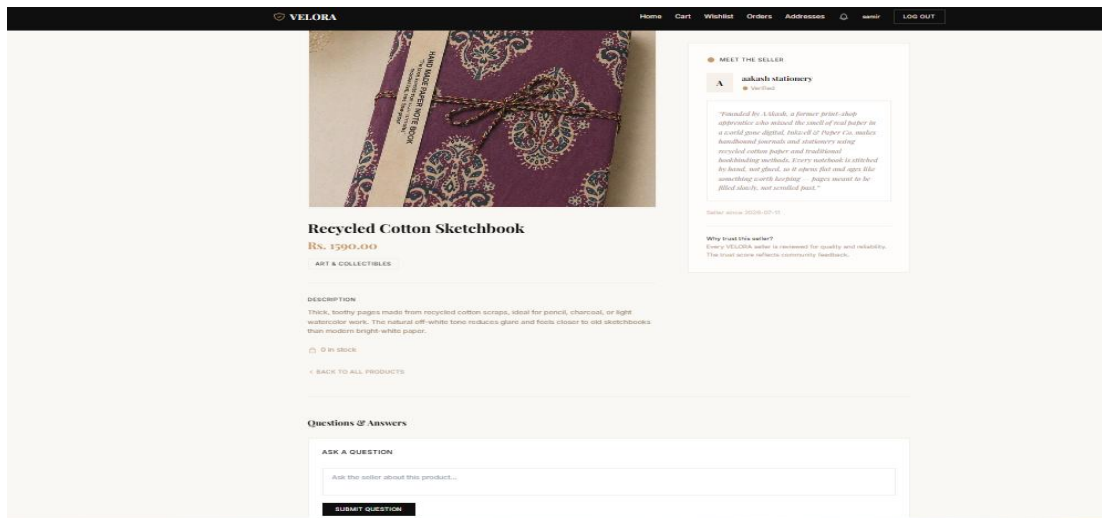
OR

 Continue with Google

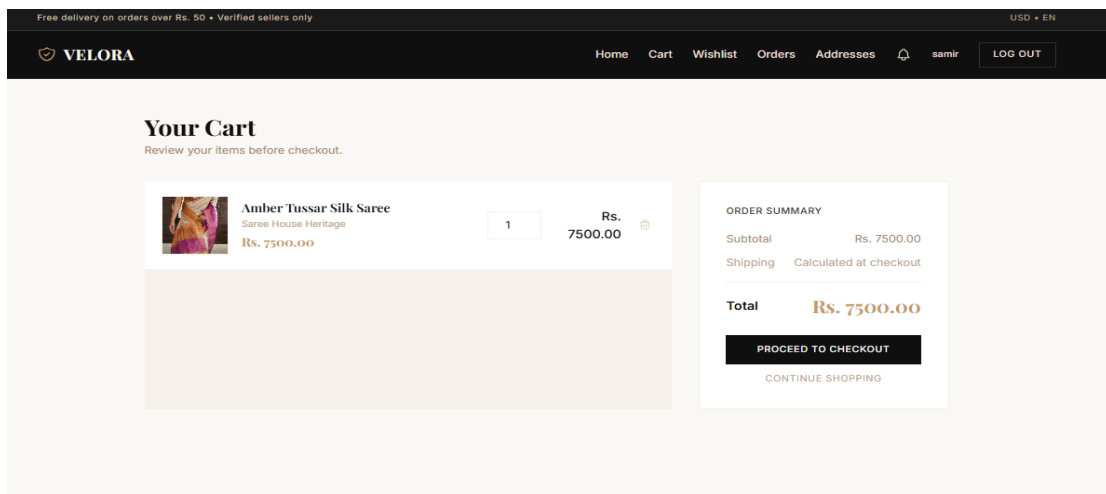
Appendix 4: Login page



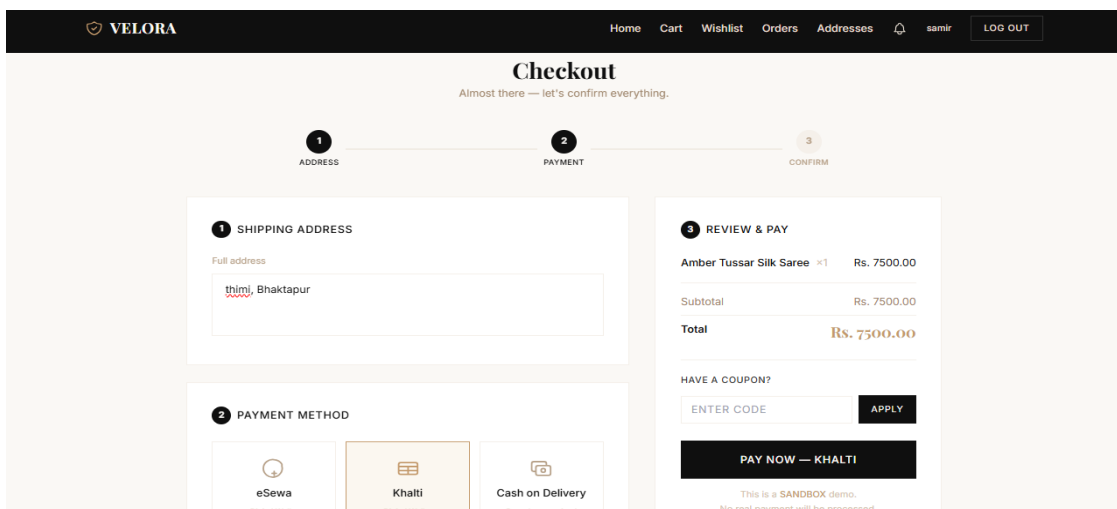
Appendix 5: Product Browsing / Search Page



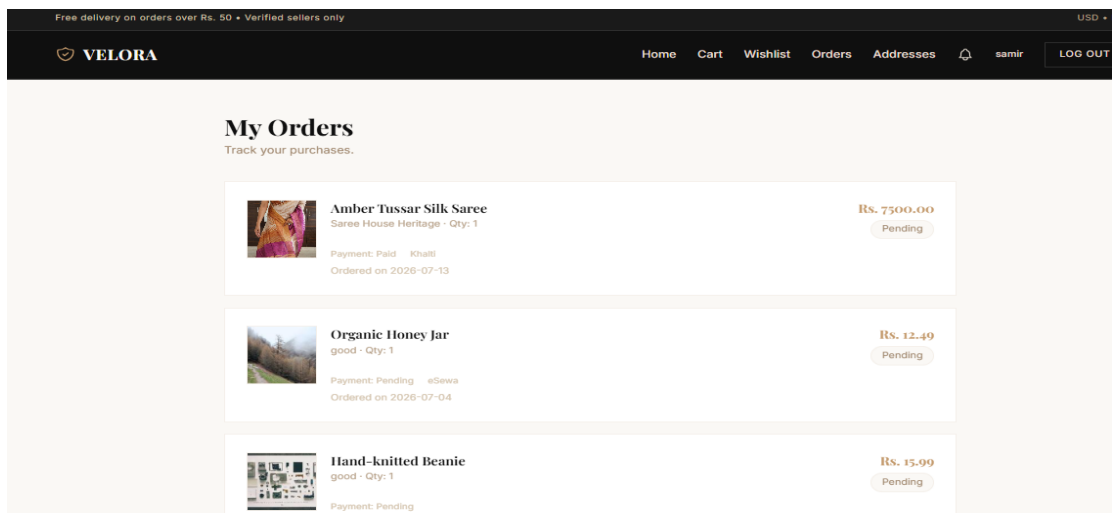
Appendix 6: Product detail page



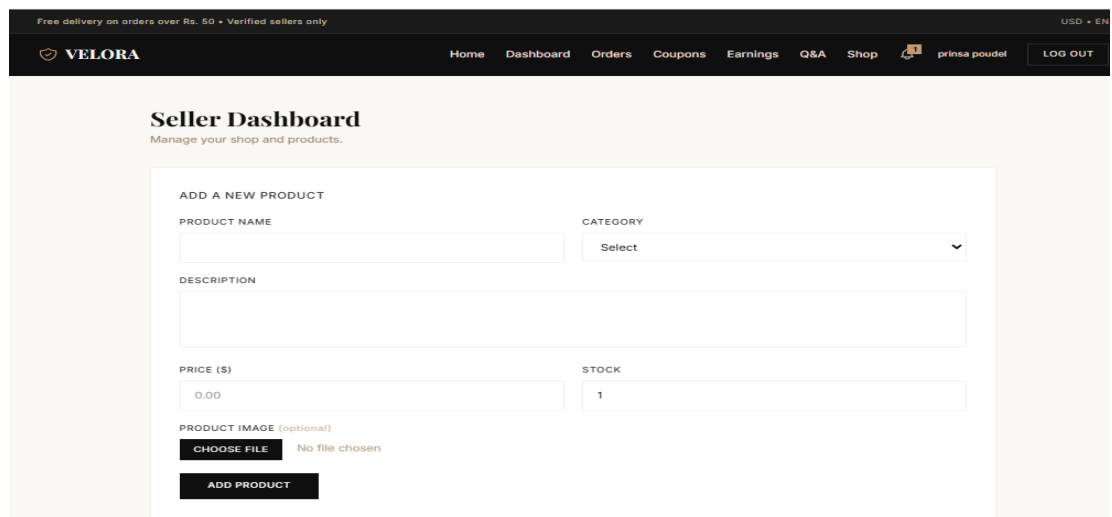
Appendix 7: Shopping Cart



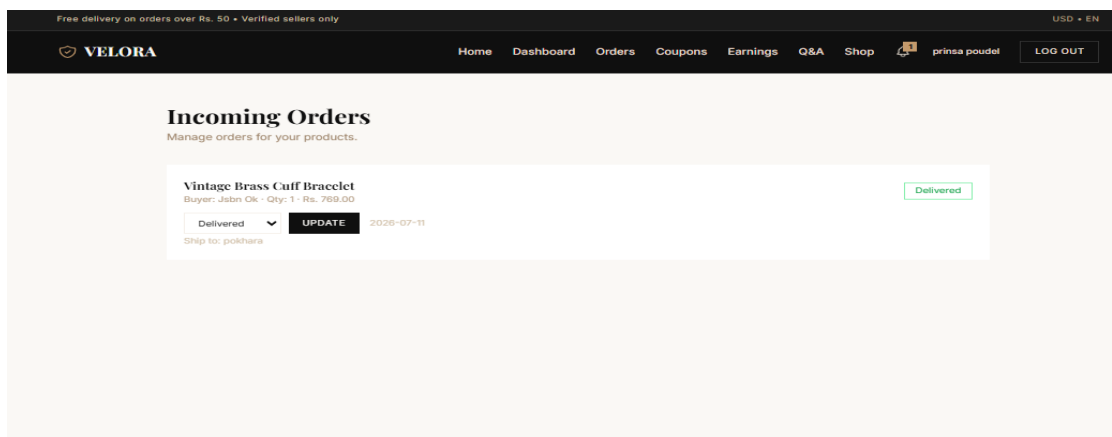
Appendix 8: Checkout / Order Placement



Appendix 9: Order tracking



Appendix 10: Seller dashboard



Appendix 11: Customer Order

Free delivery on orders over Rs. 50 • Verified sellers only

USD • EN

VELORA

HomeDashboardOrdersCouponsEarningsQ&AShopprinsa poudelLOG OUT

Earnings

Track your sales and withdraw earnings.

Rs. 769.00

GROSS SALES

-Rs. 38.45

PLATFORM FEE (5%)

Rs. 730.55

NET EARNINGS

Rs. 730.55

AVAILABLE

REQUEST WITHDRAWAL

Amount (\$)

0.00

WITHDRAW

Withdrawals are reviewed and processed by the admin team.

Appendix 12: Seller Earnings

Free delivery on orders over Rs. 50 • Verified sellers only

USD • EN

VELORA

HomeDashboardSellersProductsCouponsWithdrawalsBannersCategoriesHero ImageReportsVELORA AdminLOG OUT

Admin Dashboard

Platform overview and management.

25

TOTAL USERS

10

SELLERS

16

PRODUCTS

10

ORDERS

RS. 10000.48

TOTAL REVENUE

0

PENDING WITHDRAWALS

0

ACTIVE COUPONS

0 seller(s) pending approval

REVIEW SELLERS

Appendix 13: Admin dashboard

VELORA

HomeDashboardSellersProductsCouponsWithdrawalsBannersCategoriesHero ImageReportsVELORA AdminLOG OUT

Sellers

Approve or reject seller applications.

NAME	EMAIL	SHOP	STATUS	DOCUMENT	DATE	ACTION
kagura kagura	kagura.dreams5@gmail.com	Hearth & Clay	approved	verify_2026071101328_cloth.jpg	2026-07-11	approved
aakash magar	aakashmagar671@gmail.com	aakash stationery	approved	verify_20260711085315_Details.png	2026-07-11	approved
prinsa poudel	prinsapoudel25@gmail.com	Copper & Kohl	approved	verify_20260711084354_711pHKPjJUL_AC_UF10001000_QL80_.jpg	2026-07-11	approved
akriti	akriti2@gmail.com	Saree House Heritage	approved	verify_20260711082828_service.png	2026-07-11	approved
prinsha poudel	sadikshyaxettri5@gmail.com	skincare	approved	—	2026-07-09	approved
samira chhetri	samira2@gmail.com	Himalayan Threads	approved	—	2026-07-04	approved
Test Seller	seller@t.com	Test Shop	approved	—	2026-07-04	approved
prinsa poudel	prinsa2@gmail.com	Urban Threads	approved	—	2026-07-04	approved
jasmine	Sadikshya.design@gmail.com	fashionable	approved	—	2026-07-04	approved
sadikshya	sadikshya2@hmail.com	good	approved	—	2026-07-04	approved

Appendix 14: Seller Verification Page (admin approving a seller)

